



CHARUSAT
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

ACADEMIC REGULATIONS & SYLLABUS

(Choice Based Credit System)

Faculty of Technology & Engineering (FTE)
Chandubhai S. Patel Institute of Technology (CSPIT)



Master of Technology Programme
Thermal Engineering
(Mechanical Engineering)



**CHAROTAR UNIVERSITY OF
SCIENCE & TECHNOLOGY**

Faculty of Technology and Engineering

ACADEMIC REGULATIONS
Master of Technology (Thermal Engineering) Programme

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, Off. Nadiad-Petlad Highway, Changa 388 421, Anand,
Gujarat, INDIA Telephone: 02697-265011 & 02697-265021, Email:
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www.charusat.ac.in

Academic Year 2024-25

CHARUSAT

FACULTY OF TECHNOLOGY AND ENGINEERING

ACADEMIC REGULATIONS

Master of Technology Programmes

Choice Based Credit System

Academic regulations recommendations are provided to ensure uniform system of education, programmes duration, eligibility criteria for admission, course credits distribution, teaching and examination pedagogy, detailed syllabus with reference material.

1) System of Education

The Charotar University of Science and Technology (CHARUSAT) shall follow Choice based Credit System (CBCS) with Semester pattern at Undergraduate and Master levels. Each semester will be at least of 90 working days. Apart from the programme core courses, provision for choosing University level electives and Programme electives are available under the CBCS.

2) Duration of Programme

(i)	Undergraduate programme	(M.Tech)
	Minimum	4 semesters (2 academic years)
	Maximum	6 semesters (3 academic years)

3) Eligibility for admissions

As enacted by Government of Gujarat/AICTE/UGC from time to time.

4) Mode of admissions

As enacted by Government of Gujarat from time to time.

5) Programme Structure and Credits

As per annexure – 1 attached

6) Attendance

- 6.1. Students are expected to maintain 100% attendance in all courses. However, students may involuntarily have to miss classes due to illness or some family emergency; students are permitted to maintain a minimum attendance of 75% with producing proof or reason for the absence. In case of medical exigencies, the student/parent should inform the principal immediately through call or by email. Within a week,

starting from the day of absence, the proof of medical exigency must be submitted to the Principal's office.

- 6.2. Unauthorized absence will be considered as part of the discretionary 25% for fulfilling the minimum 75% attendance requirement for appearing in the examination.
- 6.3. Students nominated/sponsored by the University to represent in various forums like seminars/conferences/workshops/competitions or taking part in co-curricular/extra-curricular events will be given attendance credit provided the student applies in writing for such a leave in advance and obtains sanction from the Principal of his/her Institute for academic related requests.

7) Course Evaluation

7.1. The performance of every student in each course will be evaluated as follows:

- 7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment. The respective department /institute will conduct the continuous assessment. The course faculty members shall share the pedagogy related to the continuous evaluation with the students.
- 7.1.2 Final end-semester examination shall be conducted by the University through written paper, practical test, oral test, presentation by the student or a combination of these.
- 7.1.3 The weightages of continuous assessment and end-semester university examination in overall assessment shall depend on individual course as approved by Academic Council through Faculty Board and Board of Studies.
- 7.1.4 The performance of candidate in continuous assessment and in end-semester examination together (if applicable) shall be considered for deciding the final grade in a course.
- 7.1.5 In order to earn the credit in a course a student has to obtain grade other than FF.

7.2. Performance in continuous assessment and end-semester University Examination

- 7.2.1 Minimum performance with respect to continuous assessment as well as end-semester university examination will be an important consideration for passing a course.
- 7.2.2 If a candidate fails to obtain minimum required overall percentage of marks (36%), student has to repeat the examination till the minimum required overall percentage obtained.

8) Grade Point System

1. The total of the internal evaluation marks and end semester examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:
2. Proposed Grading Scheme to awarding letter grade and grade point as per NEP 2020

Letter Grade	Grade Point	Grading Scheme for Mark (In %)
O (Outstanding)	10	96.0-100
A+ (Excellent)	9	86.0-95.9
A (Very Good)	8	76.0-85.9
B+ (Good)	7	66.0-75.9
B (Above Average)	6	56.0- 65.9
C (Average)	5	46.0 – 55.9
P (Pass)	4	36.0 – 45.9
F (Fail)	0	Below 36.0
Ab (Absent)	0	Absent

➤ The minimum passing marks for each pattern of evaluation are 36%

3. The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his/her performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

(i) $SGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i

G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses in the semester

(ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i

G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

9) Award of Class

- ❖ The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	CGPA Range
First Class with Distinction	$\text{CGPA} \geq 7.0 \ \& \ \leq 10.0$
First class	$\text{CGPA} \geq 6.0 \ \& \ < 7.0$
Second Class	$\text{CGPA} \geq 5.0 \ \& \ < 6.0$
Pass Class	$\text{CGPA} < 5.0$

10) Detention Criteria

- ❖ A student will be promoted to next year only if he/she has cleared all the courses of the year he/she is studying in.

Link: <https://charusat.ac.in/> => Student's Corner => Detention Rules

**CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
(CHARUSAT)**

FACULTY OF TECHNOLOGY & ENGINEERING (FTE)

CHOICE BASED CREDIT SYSTEM

FOR

MASTER OF TECHNOLOGY

Choice Based Credit System

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist Foundation courses, Core courses, Elective courses, Non-credit (audit) courses and SWAYAM MOOCs.

1.1 Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

1.2 Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme. Environmental science will be a compulsory University core for all Undergraduate Programmes.

B. Programme Core courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

1.3 Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialised or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes. As a general guideline, Programme should incorporate 2 University Electives of 2 credits each (total 4 credits).

B. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

1.4 Non Credit Course (NC) - AUDIT Course

A 'Non Credit Course' is a course where students will receive Participation or Course Completion certificate. This will be reflected in Student's Grade Sheet but the grade of the course will not be considered to calculate SGPA and CGPA. Attendance and Course Assessment is compulsory for Non Credit Courses.

1.5 Credit Transfer through SWAYAM MOOCs

CHARUSAT provides credit transfer as per UGC guidelines to all the students from SWAYAM against elective courses. The credit transfer is offered in two modes: (a) Partial credit transfer (b) Full credit transfer.

1.6 Medium of Instruction

The Medium of Instruction will be English.

In consonance with the National Education Policy (NEP) 2020 and the guidelines of the University Grants Commission (UGC), Charotar University of Science and Technology (CHARUSAT) implements the Multiple Exit scheme in their Master of Technology programme. Table 1 shows the exit qualifications along with credit requirements.

Table 1 Exit Qualifications along with Credit Requirements

ACADEMIC LEVEL	EXIT QUALIFICATION AND CREDITS REQUIRED	NATIONAL CREDIT LEVEL (NCrF)
1st year of PG Degree	Post Graduate Diploma will be awarded after completion of 1 st year of 2-year PG program <i>Minimum of 40 credits for individuals who have completed a bachelor's programme</i>	6.5

Exit after First Year:

Award: PG Diploma in XXXX

The student shall be awarded with “PG Diploma in XXXX”, with redemption of credits from ABC. Total credits redemption shall be 1st year credits.

Department of Mechanical Engineering – Chandubhai S Patel Institute of Technology

Vision

To be recognized as a center for outstanding education and research in field of mechanical engineering.

Mission

To produce well qualified engineers, who are innovative, contributors to their profession by catering to diverse societal needs.

Program Educational Objectives:

1. To prepare graduates having sound technical knowledge and core competency in the field of thermal engineering.
2. To prepare graduates having effective communication, leadership, problem-solving, and decision making skills, supported by analytical, research and design skills to expand their knowledge; contributing to their overall personality, career development and life-long learning.
3. To prepare graduates exhibiting professional and ethical responsibilities towards their peers, employers, and society.

Program Outcomes:

PO1: An ability to independently carry out research /investigation and development work to solve practical problems.

PO2: An ability to write and present a substantial technical report/document.

PO3: Students would be able to work individually and in the team for betterment of the society.

Program Specific Outcomes:

PSO1: Students will be able to demonstrate leadership, effective communication, research potential and decision-making skills to analyse and solve manufacturing engineering problems.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
TEACHING AND EXAMINATION SCHEME FOR M TECH THERMAL ENGINEERING
CHOICE BASED CREDIT SYSTEM SCHEME (2024-25)

Semester	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Theory	Practical	Total	Credit	Theory		Practical		Total
							Internal	External	Internal	External	
Sem 1	MEUC501	Advanced Fluid Dynamics	4	0	4	4	50	50	00	00	100
	MEUC502	Advanced Thermodynamics	4	0	4	4	50	50	00	00	100
	MEUC503	Advanced Heat & Mass Transfer	4	0	4	4	50	50	00	00	100
	MEUS501	Lab Practice-I	0	4	4	2	00	00	25	25	50
	MEUS502	Lab Practice-II	0	4	4	2	00	00	25	25	50
	MEUE5XX	Programme Elective-I	3	0	3	3	50	50	00	00	100
	MEUE5XX	Programme Elective-II	3	0	3	3	50	50	00	00	100
	HSUS501/ HSUS502	Academic Speaking And Presentation Skills/ Foreign Languages (FRENCH)	2		2	2	00	00	25	25	50
					28	24					650
Sem 2	MEUC504	Energy Economics & Management	3	0	3	3	50	50	00	00	100
	MEUC505	Design of Heat Exchanger	3	0	3	3	50	50	00	00	100
	MEUC506	Experimental Techniques in Mechanical Engineering	3	0	3	3	50	50	00	00	100
	MEUS503	Lab Practice-III	0	4	4	2	00	00	25	25	50
	MEUS504	Lab Practice-IV	0	4	4	2	00	00	25	25	50
	MEUA501	Seminar	0	2	2	1	00	00	25	25	50
	MEUE5XX	Programme Elective-III	3	0	3	3	50	50	00	00	100
	MEUE5XX	Programme Elective-IV	3	0	3	3	50	50	00	00	100
	HSUA501	Academic Writing	2		2	2	00	00	25	25	50
		University Elective-II	2		2	2	00	00	25	25	50
					29	24					750

Semester	Course Code	Course Title	Credit	Examination Scheme					
				Internal		External			Total
				Progress Report	Progress Seminar	Report	Seminar	Viva	
Sem 3	MEUP601	Project Preliminaries	4	25	25	--	25	25	100
	MEUP602	Project Phase – I	18	100	150	50	100	100	500
			22						600
Sem 4	MEUP603	Project Phase – II	18	100	150	50	100	100	500
			18						500
		Grand Total	88						1100

UC: University Core

Programme Electives (PEC):

Sr. No.	Course Nature	Course Code	Course Title
1	Programme Elective-I	MEUE501	Gas Dynamics
2		MEUE502	Advanced Power Generation System
3	Programme Elective-II	MEUE503	Air Conditioning System Design
4		MEUE504	Design of Solar & Wind System
5	Programme Elective-III	MEUE505	Refrigeration & Cryogenics
6		MEUE506	Steam & Gas Turbine
7	Programme Elective-IV	MEUE507	Computational Fluid Dynamics & Heat Transfer
8		MEUE508	Modelling of IC Engine
9		MEME501	Fundamentals of Combustion (MOOC)

M. Tech. (Thermal Engineering) Programme

SYLLABI **(Semester –1)**

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
 CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUC501 ADVANCED FLUID DYNAMICS

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	0	4	4
Marks	100	0	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction and Governing Equations of Fluid Flow	15
2.	Boundary Layer	15
3.	Potential Flow	10
4.	Turbulent Flow	10
5.	Flow over immersed bodies	10

Total Hours (Theory): 60

Total Hours (Lab): 00

Total Hours: 60

B. Detailed Syllabus:

- 1. Introduction and Governing Equations of Fluid Flow** **15 Hours** **25%**
 - 1.1 Definition of fluid and concept of continuum; Properties of fluids; Flow field, streamlines, path lines and streak lines; Kinematics of fluid motion; Eulerian and Lagrangian formulations. General equations of fluid motion: Reynolds transport theorem; Mass and momentum in cartesian, polar and cylindrical coordinates, Exact Solutions of Navier-Stokes Equations: Fully developed flows, parallel flow in straight channel, Couette flow, Creeping flows.

- 2. Boundary Layer Theory** **15 Hours** **25%**

Concept of boundary layer; Derivation of boundary layer equations; Boundary layer flow over a flat plate; Boundary layer flow with non-zero pressure gradient; Flow separation and drag; Karman's momentum integral equations, Karman-Pohlhausen approximate solution.

3 Potential Flow

10 Hours

17%

Introduction to inviscid flow, Uniform flow, Source and Sink, Free vortex flow, Flow around a Rankine Half-Body, Dipole, Doublet, flow around a cylinder, Flow around a cylinder with circulation, Magnus Effect.

4 Turbulent Flow

10 Hours

17%

Characteristics of turbulent flows, Concept of averaging, Reynolds equations of motion, semi-empirical theories of turbulence, Turbulent internal flows; Turbulent external flows, velocity profiles for inner, outer and overlap layers, equilibrium boundary layers.

5 Flow over immersed bodies

10 Hours

17%

Drag and Lift on the Immersed Bodies, Drag on Sphere, Drag and Lift on a cylinder, Drag and Lift on an Aerofoil, Dimensional analysis of Drag and Lift forces.

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behaviour will be observed strictly.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Apply the fundamentals of kinematics and conservation laws of fluid flow systems.

CO2: Find approximate and exact solutions of the Navier-Stokes equation.

CO3: Solve the Boundary layer equations for laminar flow

CO4: Apply Potential flow theory to describe the different practical problems.

CO5: Describe the role of turbulence in engineering applications.

CO6: Estimate the total drag and lift forces associated with bodies immersed in the fluid

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	-	-	1
CO2	1	-	-	1
CO3	2	-	-	2
CO4	2	-	-	3
CO5	2	1	-	-
CO6	2	-	-	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:**Text Books:**

1. Muralidhar and Biswas, “Advanced Engineering Fluid Mechanics”, Alpha Science International, 2005.
2. Irwin Shames, “Mechanics of Fluids”, , McGraw Hill, 2003.
3. Fox R.W., McDonald A.T , “Introduction to Fluid Mechanics”, John Wiley and Sons Inc, 1985.
4. Pijush K. Kundu, Ira M Kohen and David R. Dawaling, “Fluid Mechanics”, Fifth Edition, 2005.

Reference Books:

1. White, F.M., "Fluid Mechanics", McGraw-Hill. 1979.
2. Schlichting, H., "Boundary Layer Theory", McGraw-Hill. 1979.
3. Garde, R.J., "Turbulent Flow", Wiley Eastern Limited. 1994.
4. Ojha, C.S.P., Berndtsson, R. and Chandaramouli, P.N., “Fluid Mechanics”, Oxford University Press. 2010.

Web Material:

1. <http://www.nptel.ac.in/>

Other Material:

1. SADHNA (Engineering Science): <http://www.ias.ac.in/sadhana/>
2. www.sciencedirect.com (Experimental Thermal and Fluid Science, International Journal of Heat and Fluid Flow)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
 CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUC502 ADVANCED THERMODYNAMICS

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	–	4	4
Marks	100	–	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	First & Second Law of Thermodynamics	08
2.	Entropy	08
3.	Availability and Irreversibility	12
4.	Thermodynamic Relations	07
5.	Nonreactive Mixtures	07
6.	Performance Analysis of Thermal Systems	18

Total Hours (Theory): 60

Total Hours (Lab): 00

Total Hours: 60

B. Detailed Syllabus:

- | | | | |
|----------|---|-----------------|------------|
| 1 | First & second Law of Thermodynamics | 08 Hours | 14% |
| 1.1 | Energy Transport mechanism, The State Postulate, How to write thermodynamic problem, Thermodynamic Problem Solving Technique, Application of First Law of Thermodynamics to Closed system, Steady flow system & Unsteady flow system. | | |
| 1.2 | Second law of thermodynamics, Absolute thermodynamic temperature scale. | | |
| 2 | Entropy | 08 Hours | 14% |
| 2.1 | Entropy, Clausius Theorem, Differential entropy balance, Entropy production mechanism. | | |
| 2.2 | Entropy change to ideal gases, Application of Second law of thermodynamics to close and open system. | | |
| 3 | Availability and Irreversibility | 12 Hours | 20% |

- 3.1 Availability, Closed system availability balance, Flow availability, Availability balance for steady flow system.
- 3.2 Irreversibility to open and close system, Exergy efficiency, Second law analysis of power plant and I.C. Engine, Thermo economics.

4 Thermodynamic Relations 07 Hours 11%

- 4.1 Maxwell Relations, Mayer's equation, T-ds equation.
- 4.2 Internal Energy and enthalpy relations for real gases, Joule Thomson effect.

5 Nonreactive Mixtures 07 Hours 11%

- 5.1 Basic definitions for gas mixtures, Dalton's law, Amagat's law, Properties of mixtures of ideal gas.
- 5.2 Entropy change due to adiabatic mixing, Mixture of perfect gas at different initial pressure and temperature, Heterogeneous system of several phases.

6 Performance Analysis of Thermal Systems 18 Hours 30%

- 6.1 Performance of Refrigeration cycles: Air refrigeration cycle, VCR, VAR
- 6.2 Performance of Steam Boilers: Equivalent evaporation, boiler efficiency, Heat losses in boiler, Heat balance sheet
- 6.3 Performance of Steam Condensers: Measurement of Vacuum, Vacuum efficiency, Condenser efficiency, Cooling water requirement
- 6.4 Testing of I.C. Engines: Morse Test, engine efficiencies, Heat balance sheet

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Apply steady flow energy equation & second law of thermodynamic to different thermodynamic systems.

CO2: Perform Energy analysis of engineering systems.

CO3: Illustrate behaviour of non-reactive mixtures.

CO4: Evaluate thermal performance of different thermal systems.

CO5: Derive thermodynamic relations between different properties of real gases.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	2	1	2
CO2	2	3	2	2
CO3	1	3	1	1
CO4	2	-	1	-
CO5	1	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Cengel, “Thermodynamics”, Tata McGraw Hill Co., New Delhi, 1980.
2. Holman, “Thermodynamics”, McGraw Hill Inc., New York, 2002.
3. R. Yadav, “Fundamentals of Engineering Thermodynamics”, Central Pub House.
4. D.P. Mishra, “Fundamentals of Combustion”, Prentice Hall of India, New Delhi, 2008.
5. Kuo K.K. “Principles of Combustion” John Wiley and Sons, 2005.

Reference Books:

1. Moren M J, “Fundamentals of Engineering Thermodynamics”, John Wiley & Sons.
2. Balmer R, “Modern Engineering Thermodynamics”, Elsevier publication.
3. Arora C P, “Thermodynamics”, Tata McGraw Hill.
4. Strehlow R A., “Fundamentals of combustion” McGraw Hill Book Company, 1984.
5. Warnatz, Maas and Dibble, “Combustion”, Springer c. 1996.

Web Material:

1. <http://nptel.iitm.ac.in/video.php?courseId=1025>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
 CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUC503 ADVANCED HEAT AND MASS TRANSFER

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	–	4	4
Marks	100	–	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Basics of Heat Transfer	02
2.	Conduction	13
3.	Convection	15
4.	Radiation	13
5.	Heat Transfer and Phase Transformation	09
6.	Mass Transfer	02
7.	Micro scale energy transport	06

Total Hours (Theory): 60

Total Hours (Lab): 00

Total Hours: 60

B. Detailed Syllabus:

- 1 Basics of Heat Transfer** **02 Hours** **4%**
 Brief Introduction to different Modes of heat transfer Conduction, General heat conduction equation – Boundary conditions – Steady simplified heat transfer in Cartesian coordinates, Finned surfaces 1 D Heat transfer with internal heat generation.
- 2 Conduction** **13 Hours** **20%**
 Conduction Rate Equation, Heat Diffusion Equation, Boundary and Initial Conditions, General conduction Equation, Conduction with Heat Generation, Extended Surfaces with Uniform and Non Uniform Cross Sections, Two Dimensional Steady State Conduction: Mathematical, Graphical and Numerical Analysis of Two Dimensional Heat Conduction

Unsteady State Conduction: Lumped Parameter Analysis, Numerical Solutions, Heisler and Semi Analytical Analysis, Analysis of fins.

- | | | | |
|---|---|-----------------|------------|
| 3 | Convection | 15 Hours | 22% |
| <p>Free & forced convection, Similarity & simulation of convection heat transfer, Boundary layer theory. Turbulent flow heat transfer, Analogy between momentum & heat transfer, Heat transfer with liquid metals, Heat transfer in high velocity flow, Recent developments in the theory of turbulent heat transfer, Natural convection under different situations, Empirical relations in convection heat transfer.</p> | | | |
| 4 | Radiation | 13 Hours | 20% |
| <p>Radiation heat transfer properties, Laws of thermal radiation, Shape factors, Radiation heat transfer between black, diffuse & gray surface, Electrical network method of solving radiation problems, Radiosity approach, Gas emission & absorption, Bulk radiations.</p> | | | |
| 5 | Heat Transfer and Phase Transformation | 09 Hours | 18% |
| <p>Boiling curve – Correlations – Nusselt's theory of film condensation on a vertical plate – Assumptions & correlations of film condensation for different geometries, Heat Pipe.</p> | | | |
| 6 | Mass Transfer | 02Hours | 4% |
| <p>Introduction, Analogy between heat and mass transfer-Mass diffusion-Fick's law of diffusion-Boundary conditions-Steady mass diffusion through a wall-Mass convection-Analogy between friction, heat transfer and mass transfer coefficients-Significance of Non – Dimensional numbers.</p> | | | |
| 7 | Micro scale energy transport | 06Hours | 12% |
| <p>Introduction to energy transport at different length and time scales, types of energy carriers, Importance of interfaces and boundaries in micro/nano scale transport, Applications to electronics cooling and energy conversion, MEMS etc.</p> | | | |

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted part of continuous internal theory evaluation.

- In the lectures discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Apply fundamentals of conduction heat transfer in various applications.

CO2: Solve heat transfer problems using principles of convection heat transfer.

CO3: Apply radiation concepts to practical application.

CO4: Analyse phase transformation with heat transfer.

CO5: Develop concepts related to mass transfer processes.

CO6: Exercise heat transfer process at micro scale and associated energy transport mechanism.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	2	2	1
CO2	3	2	2	1
CO3	3	2	2	1
CO4	2	1	2	1
CO5	1	-	1	-
CO6	1	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Incropera, Dewitt, “Fundamentals of Heat and Mass Transfer”, John Wiley & Sons (Asia) Pvt. Ltd., 2011.
2. Holman, “Heat Transfer”, McGraw Hill Inc., New Delhi, 2002.
3. Sukhatme, S.P., Heat Transfer, Universities Press (India), 1996.
4. YunusCengel, “Heat Transfer: A basic approach”, 1985.

Reference Books:

1. Lienhard J H, “Heat Transfer”, Phlogiston Press, 2017.
2. Frank Kreith & Mark.Bohn, “Principle of Heat Transfer”, 2001.
3. Robert Siegel, John Howell, “Thermal Radiation Heat Transfer”, CRC Press, 2015.
4. E. M. Sparrow, “Radiation Heat Transfer Augmented Edition”, Wadsworth Publishing Company, Inc., 2018.
5. Tien C L, Majumdar A & Gerner F M, “Microscale Energy Transport”, Taylor & Francis.

Web Material:

1. <http://nptel.ac.in/downloads/103105052/>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUS501 LAB PRACTICE–I

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	–	4	4	2
Marks	–	100	100	

A. Outline of the Course:

The lab practice consists of the tutorials and experiments as decided by the course Supervisors of the program core courses namely Advanced Heat & Mass Transfer, Advanced Thermodynamics and Combustion, Design of Solar and Wind System and Lab view software.

Total Hours (Lab): 60

Total Hours: 60

B. Instructional Method and Pedagogy:

- Attendance is compulsory in Laboratory
- Laboratory study involves demonstration/hands-on practice in relevance of course content.
- Experiments/Tutorials related to course will be carried out in the laboratory
- In the laboratory discipline and behavior will be observed strictly.

C. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Determine the thermal conductivity of Solid and Liquid.

CO2: Use the Thermal Image Camera.

CO3: Analyse the performance of Flat plate Solar Collector and Solar Still Plant.

CO4: Explain the performance various Renewable Energy systems.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	1	-	2
CO2	1	1	-	2
CO3	2	1	1	2
CO4	1	1	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:**Text Books:**

1. Incropera, Dewitt, “Fundamentals of Heat and Mass Transfer”, John Wiley & Sons (Asia) Pvt. Ltd., 2011.
2. Holman, “Heat Transfer”, McGraw Hill Inc., New Delhi, 2002.
3. P. Mishra, “Fundamentals of Combustion, Prentice” Hall of India, New Delhi, 2008.
4. Da Rosa .A.V, “Fundamentals of Renewable Energy Processes”, 2nd ed., Academic Press, 2009.

Reference Books:

1. Sen .Z, “Solar Energy Fundamentals and Modeling Techniques”, Turkey, 2008
2. Strehlow R A., “Fundamentals of combustion” McGraw Hill Book Company, 1984.
3. Robert Siegel, ,John Howell“Thermal Radiation Heat Transfer”, CRC Press, 2015.
4. E. M. Sparrow , “Radiation Heat Transfer Augmented Edition”, Wadsworth Publishing Company, Inc.,2018

Web Material:

1. <https://nptel.ac.in/courses/112/105/112105051/>

Software:

1. ANSYS Fluent
2. Solid works

Web Material:

1. <http://www.automationstudio.com/>
2. <https://www.ansys.com/>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUS502 LAB PRACTICE-II

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	–	4	4	2
Marks	–	100	100	

A. Outline of the Course:

The lab practice consists of the tutorials and experiments as decided by the course Supervisors of the program core courses namely Advanced Fluid Dynamics, Gas Dynamics, Advanced Power Generation System, and Software practice (Solid works and ANSYS).

Total Hours (Lab): 60

Total Hours: 60

B. Instructional Method and Pedagogy:

- Attendance is compulsory in Laboratory
- Laboratory study involves demonstration/hands-on practice in relevance of course content.
- Experiments/Tutorials related to course will be carried out in the laboratory
- In the laboratory discipline and behavior will be observed strictly.

C. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Find the Aerodynamic Properties of various models.

CO2: Synthesis and characterization of Advanced Heat Transfer Fluids.

CO3: Evaluate the performance of Gas Turbine Power plant.

CO4: Use the Reverse Engineering software.

CO5: Use 3D printer for real-life model.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	1	-	2
CO2	2	1	-	2
CO3	1	1	-	1
CO4	1	-	-	-
CO5	1	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:**Text Books:**

1. Muralidhar and Biswas, Advanced Engineering Fluid Mechanics, , Alpha Science International, 2005.
2. Yahya S. M., “Fundamentals of Compressible Flow”, New AGE International Limited, 2005
3. P.K. Nag, “ Power Plant Engineering”, McGraw-Hill Education.
4. Schlichting, H., "Boundary Layer Theory", McGraw-Hill. 1979.

Reference Books:

1. R.W.Haywood, “Analysis of Engineering Cycles”, 4th Edition, Pergamon Press, Oxford, 1991.
2. White, F.M., "Fluid Mechanics", McGraw-Hill. 1979.
3. Charles L. Ladson, Langley Research Center, “Computer program to obtain ordinates for NACA airfoils”, National Aeronautics and Space Administration, Langley Research Center, 1996.
4. Shapiro A. H, “Dynamics and Thermodynamics of Compressible Fluid Flow, Vols. I and II”, Ronald Press, New York, 1953.

Web Material:

1. <http://www.automationstudio.com/>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
 CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUE501 GAS DYNAMICS

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	0	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction	05
2.	Isentropic Flow with Variable Area	10
3.	Normal Shock Waves	07
4.	Oblique Shock	06
5.	Flow in Constant Area Ducts with Friction	07
6.	Flow in Constant Area Ducts with Heat Transfer	05
7.	Study of Wind Tunnels	05

Total Hours (Theory): 45

Total Hours (Lab): 00

Total Hours: 45

B. Detailed Syllabus:

- 1. Introduction** **05 Hours** **11%**
 - 1.1 Thermodynamic concept, Types of flows, Concepts of continuum and control volume.
 - 1.2 Generalized continuity, Momentum and Energy Equations.
 - 1.3 Speed of sound wave, Mach number, Physical difference between incompressible, Subsonic and supersonic flows, three reference speeds, dimensionless velocity, concepts of static and stagnation parameters.

- 2 Isentropic Flow with Variable Area** **10 Hours** **22%**
 - 2.1 Concept of Isentropic and adiabatic processes. Mach Number variation, choking in isentropic flow.

2.2 Operation of nozzles and diffusers under varying pressure ratios, performance of real nozzles, applications of isentropic flow.

3 Normal Shock Waves 7 Hours 16%

3.1 Development of a shock wave, governing equations, Rankine – Huguenot, Prandtl and other relations, weak shocks, thickness of shocks, normal shocks in ducts.

3.2 Performance of convergent–divergent nozzle with shocks, moving shock waves, shock problems in one dimensional supersonic diffuser, supersonic pitot tube.

4 Oblique Shock 06 Hours 13%

4.1 Oblique shock, governing physical equations and general relations

4.2 Shock polar diagram and auxiliary diagrams

4.3 Strong and weak shocks, detached shock, interaction and reflection of shocks.

5 Flow in Constant Area Ducts with Friction 07 Hours 16%

5.1 Fanno curves, Fanno flow equations and its solution, Fanno Flow tables.

5.2 Variation of flow properties, performance of long ducts.

5.3 Isothermal flow with friction.

6 Flow in Constant Area Ducts with Heat Transfer 05 Hours 11%

6.1 Rayleigh line, Rayleigh flow relations, Tables for Rayleigh flow.

6.2 Variation of flow properties, choking effects, shock waves with changes in stagnation temperature.

7 Study of Wind Tunnels 05 Hours 11%

7.1 Types of wind tunnels

7.2 Various flow visualization methods

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi–media projector, black board etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Apply the basic relations in thermodynamics and fluid mechanics required for compressible flow analysis and evaluate the one dimensional steady isentropic compressible flow through variable area duct.

CO2: Analyze one dimensional steady compressible flow through variable area duct with normal shock and oblique shock.

CO3: Evaluate one -dimensional steady compressible flow through a constant area duct with consideration of friction.

CO4: Evaluate one -dimensional steady compressible flow through a constant area duct with consideration of heat transfer.

CO5: Analyze Aerodynamics properties of various models with the help of wind tunnel and understand the type of compressible flow by using various flow visualization technics.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	1	-	2
CO2	2	1	-	1
CO3	2	-	-	1
CO4	2	-	-	1
CO5	1	1	1	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Yahya S. M., “Fundamentals of Compressible Flow”, New AGE International Limited, 2005.
2. Anderson, J.D., “Modern Compressible Flow: With Historical Perspective”, McGraw Hill, 2002.
3. P. Balachandran, “Fundamentals of Compressible Fluid Dynamics”, PHI Learning Pvt. Ltd., 2006.

Reference Books:

1. Rathakrishnan. E, “Gas Dynamics”, Prentice, Hall of India Pvt. Ltd., 2001
2. Yahya. S. M., “Gas Tables for Compressible Flow Calculations” New AGE International Limited, 2004.
- 3.. Shapiro A. H, “Dynamics and Thermodynamics of Compressible Fluid Flow, Vols. I and II”, Ronald Press, New York, 1953.

Web Material:

1. <http://nptel.ac.in/courses/112103021>

Other Material:

1. SADHNA (Engineering Science): <http://www.ias.ac.in/sadhana/>
2. www.sciencedirect.com (Journal of Fluid Mechanics, Journal of the Aeronautical Sciences)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
 CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUE502 ADVANCED POWER GENERATION SYSTEM

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	0	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Energy Storage	06
2.	Steam Power Plant	10
3.	Gas Turbine Power Plant	08
4.	Nuclear Power Plant	12
5.	Solar, Wind and Hydro Power Plants	09

Total hours (Theory): 45

Total hours (Practical): 00

Total hours: 45

B. Detailed Syllabus:			
1	Energy Storage	06 Hours	13%
1.1	Introduction, Thermal energy storage,		
1.2	Electrochemical energy storage		
1.3	Chemical Energy storage		
1.4	Magnetic Energy storage.		
2	Steam Power Plant	10 Hours	22%
2.1	Introduction. General layout of steam power plant, Modern coal. Fired Steam, Steam power plant. Power plant cycle, Fuel Handling, Combustion equipment, Ash handling, Dust collectors.		

3	Gas Turbine Power Plant	08 Hours	18%
3.1	Cogeneration. Combined cycle power plant, Analysis, Waste heat recovery, IGCC power plant, Fluidized bed, Combustion, Advantages and Disadvantages.		
4	Nuclear Power Plant	12 Hours	27%
4.1	Nuclear physics, Nuclear Reactor, Classification, Types of reactors, Site selection.		
4.2	Application of nuclear power plant. Nuclear Power Plant Safety: Bi-Product of nuclear power generation.		
4.3	Economics of nuclear power plant, Nuclear power plant in India, Future of nuclear power.		
5	Solar, Wind and Hydro Power Plants	09 Hours	20%
5.1	Solar Thermal Power Generation. Concentrating solar power (CSP) technologies stand alone and grid connected systems		
5.2	Wind Energy Conversion Systems; Sources and Potential, Types of wind turbines and their characteristics, Wind Characteristics.		
5.3	Hydro power plants - turbine characteristics.		

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behaviour will be observed strictly.
- Industrial visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of the assignment.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Analyze the performance of steam and gas power plants.

CO2: Identify the parameters affecting the performance of the nuclear power plant.

CO3: Describe the working of energy storage system, Solar, Wind and Hydro Power Plants.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	1	-	2
CO2	2	1	-	2
CO3	2	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E Recommended Study Material:**Text Books/ Reference Book:**

1. P.K. Nag, “ Power Plant Engineering”, McGraw-Hill Education
2. B. Banerjee, N. Sharma, “Nuclear Power In India - A Critical History”, RUPA
3. R.W.Haywood, “Analysis of Engineering Cycles”, 4th Edition, Pergamon Press, Oxford, 1991.
4. Manoj Kumar Gupta, “ Power Plant Engineering”, PHI
5. S. Sumathi, L. Ashok Kumar, P. Surekha, “Solar PV and Wind Energy Conversion Systems”, Springer Nature; 2015 edition (20 April 2015).
6. Arora S.C and Domkundwar S, “A Course in Power Plant Engineering”, DhanpatRai, 2001

Reading Materials, web materials with full citations:

1. http://nptel.iitm.ac.in/courses/Webcoursecontents/IIT%20Kharagpur/Manuf%20Proc%20II/New_index1.html
2. <http://www.sciencedirect.com/science/journal>
3. <http://www.ieindia.org/publish/pr/pr.htm>
4. <http://www.ieindia.info/public.asp/me/me.htm>
5. <http://www.ias.ac.in/sadhana>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
 CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUE503 AIR CONDITIONING SYSTEM DESIGN

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	–	3	3
Marks	100	–	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Psychrometry & Comfort Condition	07
2.	Cooling Load Calculation	08
3.	Air Conditioning Systems	06
4.	Duct Design, Air Distribution & Noise Control	10
5.	Air Conditioning Equipments & Control Systems	06
6.	Charging, Testing & Maintenance of Air Conditioning System	08

Total Hours (Theory): 45

Total Hours (Lab): 00

Total Hours: 45

B. Detailed Syllabus:

- | | | | |
|----------|--|-----------------|------------|
| 1 | Psychrometry& Comfort Conditions | 07 Hours | 18% |
| 1.1 | Psychrometry & Psychrometric properties, Use of Psychrometric chart, Different psychrometric processes. | | |
| 1.2 | Comfort chart, Factors affecting optimum effective temperature. | | |
| 2 | Cooling Load Calculation | 08 Hours | 18% |
| 2.1 | Heat flow due to heat conduction, Solar heat load, Load from occupants, Equipment load, Infiltration air load, Miscellaneous heat sources, Fresh air load. | | |
| 2.2 | Design of Air conditioning system, By-pass factor consideration, Effective sensible heat factor. | | |

- | | | | |
|----------|---|-----------------|------------|
| 3 | Air Conditioning Systems | 06 Hours | 13% |
| 3.1 | Factors affecting system selection, Different types of air conditioning system, All air system, All water system, Air water system, DX system, Evaporative cooling system, Split Air Conditioning system, Air Handling Units, VRF and VRV system. | | |
| 4 | Duct Design, Air Distribution & Noise Control | 10 Hours | 22% |
| 4.1 | Air duct, Duct design, Pressure loss, Friction loss, Use of friction chart, Dynamic losses, Losses due to area changes, Methods of duct design, Duct material, Duct work, Duct lay out consideration. | | |
| 4.2 | Selection of Air outlets, Selection procedure with the help of Nomograph, Selection procedure with tables. | | |
| 4.3 | Acoustic design, acoustic Terminology, Acoustical design criteria, Noise control, Sound Transfer, Methods of noise reduction. | | |
| 5 | Air Cleaning, Handling Equipments & Control Systems | 06 Hours | 13% |
| 5.1 | Air filter, Selection of Air filters. | | |
| 5.2 | Types of fan and blowers. | | |
| 5.3 | Controls, Control of outdoor air quantity, Fan control, Control of cooling coils, Control of humidification. | | |
| 6 | Charging, Testing & Maintenance of Air Conditioning System | 05 Hours | 12% |
| 6.1 | Installation of the Chillers, AHU, FCU, Packaged Units Condensing Units, Charging of the unit, Testing for leakage, Causes for faults and remedies. | | |

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Design air-conditioning system with by-pass factor consideration.

CO2: Calculate cooling load of given premises.

CO3: Design duct system & Distribution system.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	2	1	2
CO2	1	2	2	1
CO3	2	3	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. ASHRAE Handbook
2. “Handbook of air-conditioning system design”, Carrier Incorporation, McGraw Hill Book Co.
3. “Refrigeration & Air Conditioning for engineers”, P.S. Desai, Khanna Publishers.

Reference Books:

1. Norman C. Harris, “Modern Air Conditioning”, New York, McGraw-Hill.
2. Manohar Prasad, “Refrigeration & Air Conditioning”, New Age Publishers.
3. Jones W.P., “Air Conditioning Engineering”, Edward Arnold Publishers Ltd.
4. W.F. Stocker and J. W. Jones, “Refrigeration and Air Conditioning”, McGraw-Hill
5. Jordon and Prister, “Refrigeration and Air Conditioning”, Prentice Hall of India Pvt. Ltd.

Web Material:

1. <http://www.nptelvideos.in/2012/12/refrigeration-and-airconditioning.html>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
 CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUE504 DESIGN OF SOLAR & WIND SYSTEM

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	0	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Design Concepts of Solar Systems	07
2.	Solar Heating and Cooling Systems	15
3.	Solar Thermal Energy Storage	15
4.	Wind Energy	08

Total hours (Theory): 45

Total hours (Practical): 00

Total hours: 45

B. Detailed Syllabus:

- | | | | |
|--------------|--|---------------------|----------------|
| 1 | Design Concepts of Solar Systems | 07 Hours | 16% |
| 1.1 | System conceptual design, design of major components, overall system, design of physical principles to the solar system based on application. | | |
| 1.2 | The process includes idea generation, concepts election and estimation, design of major components, and overall system design, solar radiation data. | | |
|
2 |
Solar Heating and Cooling Systems |
15 Hours |
33% |
| 2.1 | Design of solar thermal systems for water, space heating, cooling and power generation. | | |
| 2.2 | Design of non-focusing and focusing collectors. | | |

3 Solar Thermal Energy Storage 15 Hours 33%

- 3.1 Design aspects of solar thermal energy storage systems.
- 3.2 Selection criteria of storage materials for heating and cooling applications, selection of heat transfer fluid for heating and cooling applications.
- 3.3 Design of LHTES for solar process heating and power generation applications.

4 Wind Energy 08 Hours 18%

- 4.1 Wind data and energy estimation – Wind energy Conversion systems – Wind energy generators and performance.

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behaviour will be observed strictly.
- Industrial visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of the assignment.

D. Course Outcomes (COs):

At the end of the course students will be able to:

CO1: Design the Wind turbine

CO2: Analyze the performance of various thermal systems

CO3: Design and development of solar thermal systems

CO4: Evaluate performance of solar thermal collectors.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	3	-	-	1
CO2	2	1	1	1
CO3	3	1	-	2
CO4	3	-	1	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E Recommended Study Material:

Text Books/ Reference Book:

1. Da Rosa .A.V, “Fundamentals of Renewable Energy Processes”, 2nd ed., Academic Press, 2009.
2. Duffie .J.A and Beckman .W.A, “Solar Engineering of Thermal Process”, Wiley,3rd ed., 2006.
3. Kalogirou .S.A, “Solar Energy Engineering: Processes and Systems”, Academic Press, 2009.
4. Sen .Z, “Solar Energy Fundamentals and Modeling Techniques”, Turkey, 2008.
5. Dincer .I, Rosen .M, “Thermal Energy Storage”, 2nd ed., Wiley, 2011.
6. Chauhan .D.S, Srivastava .S.K, “Non-Conventional Energy Resources”, New Age.

Reading Materials, web materials with full citations:

1. <http://nptel.ac.in>
2. <http://www.sciencedirect.com>
3. SADHNA (Engineering Science): <http://www.ias.ac.in/sadhana/>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
 CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
HSUS502 : FOREIGN LANGUAGES (French)

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	--	02	30	02
Marks	--	100	100	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to French Language	08
2.	Grammar: Articles, Tense, Forms, Numbers, Verbs, Days, Months, Family	08
3.	Grammar : Adjectives, Adverbs, Interrogative Forms, Directions, Countries, Nationalities, Seasons, Weather, Professions, Verbs	08
4.	Grammar: Prepositions, Conjunctions, Tenses, Colours, Vegetables, Fruits, Shapes, Verbs	06

Total hours (Theory): 00

Total hours (Practical): 30

Total hours: 30

B Detailed Syllabus:

1.	Introduction to French Language	08 Hours	28%
	Facts and figures about French Language; Basic French Linguistics -* Alphabets * Accents * Liaison * Nasalization French Culture, Differ between French and English; Grammar -Subject Pronoun, Verbs: (être, avoir, habiter, regarder, manger ... “er” verb), Form of address, Numbers (1 to 20), Nouns and plurals of nouns, The expression: C’est, Il y a; Presentation : -1) Self-Introduction-2) Question and answering; Dialogue		
2.	Grammar: Articles, Tense, Forms, Numbers, Verbs, Days, Months, Family	08 Hours	28%
	Grammar -Definite articles, Indefinite articles, Present tense (Positive Forms, Negative Forms), Numbers (21 to 100, 100-1000), Days, Months, Family, Verbs: (aller, venir, finir, pouvoir,		

	vouloir ... “ir” verb); Social Links -1), My family & relations 2) Appointments 3) Gathering information from someone; Dialogue		
3.	Grammar : Adjectives, Adverbs, Interrogative Forms, Directions, Countries, Nationalities, Seasons, Weather, Professions, Verbs	08 Hours	28%
	Grammar - Common Adjectives, Comparative Adjectives, Common Adverbs, Interrogative Forms, The expression: “On”, Directions, Countries, Nationalities, Seasons, Weather, Professions, Verbs: (Prendre, Apprendre, Comprendre, faire ... “re “ verb); Work , Study and Travel -1) Job/ Profession 2) Ticket Reservation (At Bus/At Railway/At Airport); Dialogue		
4.	Grammar: Prepositions, Conjunctions, Tenses, Colours, Vegetables, Fruits, Shapes, Verbs	06 Hours	26%
	Grammar -1) Common Prepositions 2) Common Conjunctions 3)Past Tense 4) Future Tense 5) Colors ,Shapes, Animals ,Vegetables, Fruits 6) Verbs: (“er”, “ir”, ”re” etc...); Food & Shopping -1) Buy a vegetables and fruits 2) Any Conversation between Customer and Vendor (At Mall/At Restaurant / At Market); Dialogue		

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.
- Material testing laboratories visits will be organized for students to explore the facilities. Students are required to prepare a report on visit and submit as a part of the assignment.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Gain basic communication skills in French language with preliminary understanding of grammar.

CO2: Develop vocabulary required to speak about him/herself and his/her immediate environment.

CO3: Become capable of interacting in simple ways, to ask simple questions to get necessary information, to reply simple questions.

CO4: Become capable of understanding and using simple instructions in their personal, academic and professional environments.

CO5: Develop skills and intelligences to function in multi-disciplinary and cross-cultural work environment.

CO6: Practice new global trends in communication in multiple perspectives at personal, professional, and social level.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	-	-	2	-
CO2	-	-	2	-
CO3	-	-	2	-
CO4	-	-	2	-
CO5	-	-	2	-
CO6	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E Recommended Study Material:

Text Books/ Reference Book:

1. Complete French: All-In-One, McGraw-Hill, Amazon
2. Best for Grammar: Easy French Step-by-Step, McGraw-Hill, Amazon
3. Basic French: McGraw-Hill, Amazon
4. French Grammar for Beginners, Amazon

Reading Materials, web materials with full citations:

1. <https://alison.com/course/french-language-studies-introduction>
2. <https://alison.com/course/basic-french-language-skills-for-everyday-life-revised-2017>
3. <http://www.bbc.co.uk/languages/french/>
4. <https://www.loecsen.com/en/learn-french>
5. <https://www.youtube.com/watch?v=ujDtm0hZyII>
6. <https://alison.com/course/french-language-studies-introduction>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
 CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
 HSUS501 : ACADEMIC SPEAKING AND PRESENTATION SKILLS

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	--	02	30	02
Marks	--	100	100	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Foundations of Advance Communication	04
2.	Art of Conversation	06
3.	Science of Power Speaking	06
4.	Academic Speaking Application – Part I	08
5.	Academic Speaking Application – Part II	06

Total hours (Theory): 00

Total hours (Practical): 30

Total hours: 30

B Detailed Syllabus:

1.	Foundations of Advance Communication	04 Hours	14%
	Meaning and Definition of Advance Communication; Advance Communication in Digital, Social, Mobile World; Strategies for Advance Communication; Meaning and Concept of Academic Language; High Frequency Academic Vocabulary		
2.	Art of Conversation	06 Hours	20%
	Describing people, places and things; Expressing opinions; Making suggestions; Persuading someone; Interpreting and Summarizing		
3.	Science of Power Speaking	06 Hours	20%
	Phonemes, Word Stress, Pronunciation, Intonation, Pause, Register, Fluency, Prosody, Lexical Range		
4.	Academic Speaking Application – Part I	08 Hours	26%

	Art of Oratory, Formal Presentation, Speech Analysis – Decoding Best Speeches		
5.	Academic Speaking Application – Part II	06 Hours	20%
	Job Interview, Group Discussion, Meeting		

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.
- Material testing laboratories visits will be organized for students to explore the facilities. Students are required to prepare a report on visit and submit as a part of the assignment.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Understand and demonstrate advance communication and academic speaking skills

CO2: Demonstrate ability to communicate in diverse situations

CO3: Activate and extend their linguistic and communicative competence

CO4: Demonstrate the formal presentation skills

CO5: Demonstrate performing ability at group discussion and personal interview

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	-	-	2	-
CO2	-	-	2	-
CO3	-	-	2	-
CO4	-	-	1	-
CO5	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E Recommended Study Material:

Text Books/ Reference Book:

1. Business Communication Today (Thirteenth Edition) by Courtland L. Bovee, John V. Thill and Roshan Lal Raina
2. Effective Speaking Skills by Terry O' Brien
3. Speak Better Write Better by Norman Lewis
4. Well Spoken: Teaching Speaking to All Students by Erik Palmer
5. Let Us Hear Them Speak : Developing Speaking – Listening Skills in English by Jayshree Mohanraj (Publisher – Sage Publication)
6. The craft of scientific presentations: Critical steps to succeed and critical errors to avoid. New York: Springer by Michael Alley
7. Presentation Skills in English by Bob Dignen (Publisher: Orient Black Swan)

Reading Materials, web materials with full citations:

1. TED Talk : How to speak so that people want to listen
https://www.ted.com/talks/julian_treasure_how_to_speak_so_that_people_want_to_listen?language=en
2. TED Talk: The 110 techniques of communication and public speaking
https://www.ted.com/talks/david_jp_phillips_the_110_techniques_of_communication_and_public_speaking

M. Tech. (Thermal Engineering) Programme

SYLLABI **(Semester –2)**

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CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUC504 ENERGY ECONOMICS AND MANAGEMENT

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	0	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	An overview of energy management.	05
2.	Energy auditing.	10
3.	Energy conservation opportunities.	10
4.	Economic Analysis	05
5	Sustainable Growth through Policy supports	15

Total Hours (Theory): 45

Total Hours (Lab): 00

Total Hours: 45

B. Detailed Syllabus:

- | | | |
|--|-----------------|------------|
| 1. An overview of energy management | 05 Hours | 11% |
| 1.1 Energy Flow Diagram | | |
| 1.2 Global Trends in Energy Use | | |
| 1.3 India and World- Supply and demand | | |
| 1.4 Energy and Environment | | |
| 1.5 Energy and Quality of Life | | |
| 2 Energy auditing. | 10 Hours | 22% |
| 2.1 Energy Audit: Meaning. | | |
| 2.2 Energy Audit Methodology | | |
| 2.3 Energy Audit Reporting Format | | |

3	Energy conservation opportunities.	10 Hours	22%
3.1	Energy conservation opportunities in Boiler and fired systems		
3.2	Energy conservation opportunities in steam and condensate system		
3.3	Energy conservation opportunities in Boiler and fired systems		
3.4	Cogeneration		
4	Economic Analysis	10 Hours	22%
4.1	Economic factors		
4.2	Simple Payback Period,		
4.3	Time Value of Money- discount rate,		
4.4	Criteria for Assessing Energy Projects –(Net Present Value (NPV),		
4.5	Benefit/Cost Ratio (B/C),		
	Inflation, Internal Rate of Return (IRR).		
5	Sustainable Growth through Policy supports	10 Hours	22%
5.1	International, national laws and policy adopted or being adopted.		
5.2	Demand side management in the current scenario.		
5.3	Typical Format of an Energy Policy		

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behavior will be observed strictly.

D Course Outcomes (COs):

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At the end of the course, the students will be able to

CO1: Criticize the Energy management scenario.

CO2: Capable to Prepare all energy audit.

CO3: Identify energy consumption opportunities

CO4: Suggest and plan energy conservation solutions.

CO5: Predict the future trend of energy.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	1	1	1	-
CO2	2	2	1	2
CO3	1	1	3	1
CO4	1	2	3	2
CO5	1	1	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:**Text Books:**

1. Bala Bhaskar, “Energy Security and Economic Development in India”, TERI press, 2012.
2. Frank Kreith, D. Yogi Goswami, “Energy Management and Conservation Handbook Mechanical and Aerospace Engineering Series”, CRC press, 2007.
3. Kreith F, & Goswami D Y, “Energy management and conservation handbook”, CRC Press, 2007.
4. Abbi Y. P. & Jain S, “Handbook on energy audit and environment management, The Energy and Resources Institute” (TERI), 2006.
5. Abbi Y. P, “Energy audit: Thermal power, combined cycle, and cogeneration plants, The Energy and Resources Institute” (TERI), 2012.
6. Desai S, Handbook of Energy Audit, Mcgraw Higher Ed, 2015.

Reference Books:

1. D.A. Reeg, “Industrial Energy Conservation”, Pergamon Press, 1980.
2. T.L. Boyen, “Thermal Energy Recovery” Wiley, 1980.
3. W.C. Turner, “Energy Management Handbook “, Wiley, New York, 1982.
4. I.G.C. Dryden, “The Efficient Use of Energy “, Butterworth, London, 1982.

Web Material:

- 1 Energy Mnagement National Productivity Council.
- 2 <http://www.npcindia.gov.in/competencies/energy-management>.
- 3 EA/EM Guide Books (<http://www.bee-india.nic.in/index.php?module=tri&id=4>)
- 4 Publications of The Energy and Resources Institute (TERI) www.teriin.org/publications
Publications of Bureau of Energy Efficiency.

Other Material:

1. Vikaspedia (Government of India initiative): <http://vikaspedia.in/energy/energy-efficiency/>

2. <https://www.sciencedirect.com> (Energy Conversion and Management, Energy for Sustainable Development)
3. <https://www.springeropen.com/journals> (Energy Efficiency).
4. <https://www.tandfonline.com> (International Journal of Sustainable Energy).

3. Doebelin, “Measurement System”, Tata McGraw-Hill Education, 1984.
4. Beckwith, “Mechanical Measurements”, Pearson Education India, 2007.
5. Morris. A.S, “Principles of Measurements and Instrumentation”, Prentice Hall of India, 1998.

Web Material:

1. <http://www.nptel.ac.in>
2. https://onlinecourses.nptel.ac.in/noc24_hs24/preview

Other Material:

1. SADHNA (Engineering Science): <http://www.ias.ac.in/sadhana/>
2. www.sciencedirect.com

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
 CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUC505: DESIGN OF HEAT EXCHANGERS

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	–	3	3
Marks	100	–	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to heat exchangers and its classification	05
2.	Design of double pipe heat exchangers	09
3.	Design of shell & tube heat exchangers	13
4.	Compact & phase change heat exchangers	13
5.	Design of cryogenic heat exchanger	05

Total Hours (Theory): 45

Total Hours (Lab): 00

Total Hours: 45

B. Detailed Syllabus:

- 1 Introduction to heat exchangers and its classification 05 Hours 15%**
 Basic design methodologies: Classification of heat exchanger, selection of heat exchanger, Thermal-Hydraulic fundamentals, Overall heat transfer coefficient, LMTD method for heat exchanger analysis for parallel, counter, multipass and cross flow heat exchanger, e-NTU method for heat exchanger analysis, Fouling, Rating and sizing problems, heat exchanger design methodology

- 2 Design of double pipe heat exchangers 10 Hours 25%**

- 2.1 Thermal and Hydraulic design of inner tube and annulus, hairpin heat exchanger with bare and finned inner tube, total pressure drop

3 Design of shell & tube heat exchangers 13 Hours 25%

- 3.1 Basic components, basic design procedure of heat exchanger, TEMA code, J-factors, conventional designs methods, Bell-Delaware method.

4 Compact & phase change heat exchangers 13 Hours 25%

- 4.1 Heat transfer enhancement, plate fin heat exchanger, tube fin heat exchanger, heat transfer and pressure drop
- 4.2 Condensers classification and design methods for surface condensers, Evaporators Classification and design methods

5 Design of cryogenic heat exchanger 05 Hours 10%

- 5.1 Design of heat exchangers for liquefaction systems ,single tube ,double tube Linde heat exchangers three channel heat exchangers ,multiple tube type ,Giauque Hampton and Collins type heat exchangers
- 5.2 Finned tube and plate type heat exchangers, different configuration heat transfer coefficients and friction coefficient for various configurations.

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Evaluate performance parameters and rate different types of heat exchangers.

CO2: Design double pipe heat exchangers.

CO3: Design compact and shell & tube type heat exchangers.

CO4: Design various phase change and cryogenic heat exchangers.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	1	1	1
CO2	3	2	3	1
CO3	3	2	3	1
CO4	2	-	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:**Text Books:**

1. Ramesh K Shah, “Fundamentals of Heat Exchanger Design”, Wiley Publication. 2015.
2. A.P. Fraas and M.N. Ozisick, “Heat Exchanger Design”, John Wiley & sons., New York.
3. Singh K.P., & Soler, A.L., Mechanical Design of Heat Exchangers, Arcturus Pub. Inc. N.J. 08003, USA. 1984.

Reference Books:

1. Kuppan, T, Macel Dekker, “Heat Exchanger Design Handbook”, CRC Press, 2017.
2. Schunder E.U., “Heat Exchanger Design Hand Book”, Hemisphere Pub.1983.
3. Joshi, M.V., & Mahajani, V.V., Process Equipment Design, Macmillan, India, Ltd., 1996.
4. Donald Q Kern, “Process Heat transfer”, Tata McGraw Hill, 2017

Web Material:

1. <http://nptel.ac.in/courses/112101097/25>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
 CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUC506 EXPERIMENTAL TECHNIQUES IN MECHANICAL
ENGINEERING

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	0	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction	05
2.	Analysis of Experimental data	10
3.	Measurement of Thermo-Physical Properties	24
4.	Microprocessors and Computers in Measurement	06

Total Hours (Theory): 45

Total Hours (Lab): 00

Total Hours: 45

B. Detailed Syllabus:

1. **Introduction** **05 Hours** **11%**
 Role of experiments in fluid, layout of fluid flow experiments, Instrument Classification, Characteristics of Instruments – Static and dynamic.

2. **Analysis of Experimental data** **10 Hours** **22%**
 Sources of error in experiments, Experimental error analysis, Systematic and random errors, data analysis, design of experiments, uncertainty analysis, propagation of uncertainty; Statistical analysis of experimental data- normal error distributions (confidence interval and level of significance, Chauvenet's criterion), Chi-square test of goodness of fit, method of

least squares (regression analysis, correlation coefficient), multivariable regression, graphical analysis and curve fitting.

3 Measurement of Thermo-Physical Properties 24 Hours 54%

- 3.1 Measurement of displacement, force and torque and strain measurements.
- 3.2 Measurement of pressure, measurement of high & low pressures, dynamic characteristics of pressure measuring devices.
- 3.3 Measurement of temperature, temperature measurements by mechanical effects, electrical effects and by radiation, transient response of thermal systems, temperature measurement in high speed flow.
- 3.4 Motion and vibration measurements, principles of seismic instruments, practical considerations of seismic instruments.
- 3.5 Measurement of smoke, dust and moisture, Measurement of pH
- 3.6 Surface Roughness measurement, Measurement of surface defect, Compressive and tensile strength measurement, Cutting force measurement.

4 Microprocessors and Computers in Measurement 06 Hours 13%

- 4.1 Data logging and acquisition, Use of intelligent instrument for error reduction, Elements of micro-computer interfacing, Intelligent instruments in use.

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behaviour will be observed strictly.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Analyze the experimental data to do the error analysis and to find the uncertainty in the results.

CO2: Illustrate the various measurement techniques used in the experiments to measure various thermos-physical properties.

CO3: Identify the various intelligent instruments for data acquisition.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	3	-	-	2
CO2	1	2	-	1
CO3	2	1	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Holman, J.P., “Experimental methods for engineers”, McGraw-Hill, 7th Edition, 2008.
2. S P Venkateshan, “Mechanical Measurements”, Anne Books Pvt. Ltd., 2015.
3. Raman, C.S. Sharma, G.R., Mani, V.S.V., “Instrumentation Devices and Systems”, Tata McGraw-Hill, New Delhi, 1983.

Reference Books:

1. J R Taylor, “An Introduction to Error Analysis: The Study of Uncertainties in Physical Measurements”, University Science Books, 1997.
2. Yahya. S. M., “Gas Tables for Compressible Flow Calculations” New AGE International Limited, 2004.
3. Doebelin, “Measurement System”, Tata McGraw-Hill Education, 1984.
4. Beckwith, “Mechanical Measurements”, Pearson Education India, 2007.
5. Morris. A.S, “Principles of Measurements and Instrumentation”, Prentice Hall of India, 1998.

Web Material:

1. <http://www.nptel.ac.in>

Other Material:

1. SADHNA (Engineering Science): <http://www.ias.ac.in/sadhana/>
2. www.sciencedirect.com

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUS503 LAB PRACTICE–III

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	–	4	4	2
Marks	–	100	100	

A. Outline of the Course:

The lab practice consists of the tutorials and experiments as decided by the course Supervisors of the Program Core Courses namely Experimental Techniques in Thermal Engineering, Computational Fluid Dynamics and Heat Transfer.

Total Hours (Lab): 60

Total Hours: 60

B. Instructional Method and Pedagogy:

- Attendance is compulsory in Laboratory
- Laboratory study involves demonstration/hands-on practice in relevance of course content.
- Experiments/Tutorials related to course will be carried out in the laboratory
- In the laboratory discipline and behavior will be observed strictly.

C. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Simulate the Fluid flow and heat transfer problem using analysis software like Ansys.

CO2: Use different measuring instruments.

CO3: Analyze the experimental data to do the error analysis and to find the uncertainty in the results.

CO4: Do the Data analysis using statistical analysis Software like Minitab.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	3	1	1	2
CO2	2	-	-	-
CO3	2	1	-	2
CO4	2	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Holman, J.P., “Experimental methods for engineers”, McGraw-Hill, 7th Edition, 2008.
2. John D Anderson Jr., “Computational Fluid Dynamics: The Basics with Applications”, McGraw-Hill, 1995.

Reference Books:

1. J R Taylor, “An Introduction to Error Analysis: The Study of Uncertainties in Physical Measurements”, University Science Books, 1997.
2. Veersteeg and Malalasekara, “CFD: The Finite Volume Method by”, Prentice Hall, 1996.
3. Goshdastidar, “Computer Simulation of Flow and Heat Transfer”, Tata-McGraw Hill.

Software:

1. ANSYS Fluent
2. Minitab
3. COMSOL

Web Material:

1. <https://nptel.ac.in/courses/112/105/112105045/>
2. <https://nptel.ac.in/courses/112/107/112107242/>
3. <https://nptel.ac.in/courses/108/105/108105064/>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
 CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUS504 LAB PRACTICE–IV

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	–	4	4	2
Marks	–	100	100	

A. Outline of the Course:

The lab practice consists of the tutorials and experiments as decided by the course Supervisors of the Program Core Courses namely, Energy Economics and Management, Air conditioning System Design, Refrigeration and Cryogenic, Design of Heat Exchanger, Steam and Gas Turbine, Modelling of IC Engine.

Total Hours (Lab): 60

Total Hours: 60

B. Instructional Method and Pedagogy:

- Attendance is compulsory in Laboratory
- Laboratory study involves demonstration/hands-on practice in relevance of course content.
- Experiments/Tutorials related to course will be carried out in the laboratory
- In the laboratory discipline and behavior will be observed strictly.

C. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Do Energy Audit

CO2: Design, analyse and optimize different types of heat exchangers.

CO3: Analyze the performance of steam and gas turbine.

CO4: Use LABVIEW software for data acquisition in measurement.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	2	-	2
CO2	3	1	-	2
CO3	3	1	-	2
CO4	2	1	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Frank Kreith, D. Yogi Goswami, “Energy Management and Conservation Handbook Mechanical and Aerospace Engineering Series”, CRC press, 2007.
2. “Refrigeration & Air Conditioning for engineers”, P.S. Desai, Khanna Publishers.
3. Barron R., “Cryogenic Systems”, Plenum Press.
4. Ramesh K Shah, “Fundamentals of Heat Exchanger Design”, Wiley Publication. 2015.
R. K. Hegde, “Power Plant Engineering”, Pearson India Education.
5. J.I Ramos, “Internal Combustion Engine Modelling” Hemisphere Publishing Corporation, 1989.
6. Yunus Cengel, “Heat Transfer : A basic approach”, 1985.

Reference Books:

1. W.C. Turner, “Energy Management Handbook “, Wiley, New York, 1982
2. W.F. Stocker and J. W. Jones, “Refrigeration and Air Conditioning”, McGraw-Hill
3. Hastlden, C., “Cryogenic Fundamentals”, Academic Press
4. Kuppan, T, Macel Dekker, “Heat Exchanger Design Handbook”, CRC Press, 2017.
5. Joshi, M.V., & Mahajani, V.V., Process Equipment Design, Macmillan, India, Ltd., 1996.
6. Ramoss A L, "Modeling of Internal Combustion Engines Processes", McGraw Hill Publishing Co., 1992.

Software:

3. ANSYS Fluent
4. Solid works
5. COMSOL

Web Material:

1. <https://nptel.ac.in/courses/112/105/112105248/>
2. <https://nptel.ac.in/courses/112/107/112107216/>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUA501 : SEMINAR

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	1
Marks	-	50	50	

A. Outline of the Course:

- Seminar should be based on any latest topics related to the area of mechanical engineering and/or inter-disciplinary.
- It should involve exhaustive literature survey that should include books, periodicals, journals and various internet resources.
- Student should analyse the contributions in different resources and represent it in the form of seminar report in prescribed format.
- The idea behind the seminar system is to familiarize students more extensively with the methodology of their chosen subject and also to allow them to interact with examples of the practical problems that always occur during research work.

B. Instructional Methods and Pedagogy:

- Students will select related topic based on subjects they learnt and other literatures like books, periodicals, journals and various internet resources.
- Students would be taught to make presentations and report based on it.
- Faculty can suggest modifications to improve technical subject matter and presentation skills.
- Each student has to prepare a write-up of about 25 pages. The report typed on A4 sized sheets and bound in the necessary format should be submitted after approved by the guide and endorsement of the Head of Department.
- The student has to deliver a seminar talk in front of the teachers of the department and his classmates. The teacher based on the quality of work and preparation and understanding of the candidate shall do an assessment of the seminar.
- At the start of course, the course delivery pattern and prerequisites will be discussed.
- Research / technical papers in relevant areas must be covered.

C Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Identify the latest trends in engineering by critical literature review and establish scope of work.

CO2: Assimilate literature on technical articles of specified topic and develop comprehension.

CO3: Design, develop and deliver presentation on specified technical topic.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	1	-	1	1
CO2	1	-	-	1
CO3	2	3	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D Recommended Study Material:

1. IEEE : www.ieeeexplore.ieee.org
2. Elsevier : www.sciencedirect.com
3. Sadhna (Engineering Science) (<http://www.ias.ac.in/sadhana/>)
4. Programming Languages: C, C++, MATLAB
5. Software's: AutoCAD, Creo, ANSYS

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
 CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUE505 REFRIGERATION & CRYOGENICS

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	–	3	3
Marks	100	–	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	VCR & VAR Cycles	6
2.	Compound Compression	6
3.	Nonconventional Refrigeration system	6
4.	Gas Liquefaction Cycle	6
5.	Properties of Engineering Materials at Cryogenic Temperature	8
6.	Cryogenic Instrumentation	7
7.	Cryogenic Insulation	6

Total Hours (Theory): 45

Total Hours (Lab): 00

Total Hours: 45

B. Detailed Syllabus:

- 1 VCR & VAR Cycles 06 Hours 14%**
 - 1.1 Mathematical analysis of VCR system, Factors affecting performance of VCR system, Actual VCR system, Eco refrigerants, GWP & ADP for Refrigerants.
 - 1.2 Types of VAR system, Desirable properties of refrigerant and absorbent in absorption system, Elementary properties of binary mixtures, h-c diagram, Analysis of VAR system.
- 2 Compound Compression 06 Hours 14%**
 - 2.1 Multi stage compression system, Compound compression with intercooler, System with flash chamber, Two stage compression with flat inter cooler, Multiple evaporator system.

- | | | |
|----------|---|---------------------|
| 3 | Nonconventional Refrigeration system | 06 Hours 14% |
| 3.1 | Analysis of steam jet refrigeration system, Vortex tube refrigeration system, Pulse tube refrigeration system. | |
| 4 | Gas Liquefaction Cycle | 06 Hours 14% |
| 4.1 | Ideal systems, Linde, Linde dual pressure system, Claude, Heylandt, Kapitza systems, Cascade cycle. | |
| 5 | Properties of Engineering Materials at Cryogenic temperature | 08 Hours 16% |
| 5.1 | Mechanical properties, Thermal properties, Electric & magnetic properties. | |
| 5.2 | Super conducting materials, Thermo electric materials, Composite materials, Properties of Cryogenic fluids, Super fluidity of He3 & He4. | |
| 6 | Cryogenic instrumentation | 07 Hours 14% |
| 6.1 | Temperature measurements, pressure measurements, flow measurements, liquid level measurements, fluid quality measurements. | |
| 7 | Cryogenic Insulation | 07 Hours 14% |
| | Various types of cryogenic insulation such as expanded foams, gas filled& fibrous insulation, vacuum insulation, evacuated powder& fibrous insulation, opacified powder insulation, multi-layer insulation, comparison of performance of various insulations. | |

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Design various components of VCR & VAR systems.

CO2: Evaluate performance of different refrigeration systems.

CO3: Analyze performance of different non-conventional refrigeration systems.

CO4: Compare different cryogenic insulation systems.

CO5: Identify change in mechanical, thermal, electrical and magnetic properties of mechanical materials at cryogenic temperature.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	2	1	2
CO2	1	2	2	1
CO3	2	3	1	1
CO4	1	2	1	1
CO5	2	2	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. P.S. Desai, “Refrigeration & Air Conditioning for engineers”, Khanna Publishers.
2. Barron R., “Cryogenic Systems”, Plenum Press.
3. Hastlden, C., “Cryogenic Fundamentals”, Academic Press.

Reference Books:

1. Arora C.P, “Modern Air Conditioning”, McGraw-Hill.
2. Manohar Prasad, “Refrigeration & Air Conditioning”, New Age Publishers.
3. W.F. Stocker and J. W. Jones, “Refrigeration and Air Conditioning”, McGraw-Hill.
4. Jordon and Prister, “Refrigeration and Air Conditioning”, Prentice Hall of India Pvt. Ltd.

Web Material:

1. <http://www.nptelvideos.in/2012/12/refrigeration-and-airconditioning.html>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
 CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUE506 STEAM & GAS TURBINE

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	0	100	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction	02
2.	Steam Nozzles	07
3.	Steam Turbine	20
4.	Gas Turbine:	14
5.	Jet Propulsion:	02

Total hours (Theory): 45

Total hours (Practical): 00

Total hours: 45

B Detailed Syllabus:

1 Introduction to steam and gas turbine **02 Hours** **04%**

1.1 Working principle and applications of steam and gas turbine

2 Steam Nozzles **07 Hours** **16%**

2.1 Types of nozzles, velocity of steam, discharge through nozzle, critical pressure ratio and condition for maximum discharge, physical significance of critical pressure ratio, nozzle efficiency.

3 Steam Turbine **20 Hours** **45%**

3.1 Principle of operation, types of steam turbines, compounding of steam turbines,

3.2 Impulse turbine – velocity diagram, calculation of work, power and efficiency, condition for maximum efficiency,

- 3.3 Reaction turbines – velocity diagram, degree of reaction, reheat factor,
- 3.4 Governing of steam turbine – throttle, nozzle and bypass governing,
- 3.5 Methods of attachment of blades to turbine rotor, Labyrinth packing, Losses in steam turbine, Special types of steam turbine- back pressure, pass out and mixed pressure turbine.
- 3.6 Steam Turbine Design Process, Turbine Blade Construction, Materials, and Manufacture.
- 3.7 Erosion protection.

4 Gas Turbine **14 Hours** **31%**

- 4.1 Classification, open and closed cycle, gas turbine fuels, actual Brayton cycle,
- 4.2 Optimum pressure ratio for maximum thermal efficiency, work ratio, air rate, effect of operating variables on the thermal efficiency and work ratio, and air rate
- 4.3 Simple open cycle turbine with regeneration, reheating and Intercooling, Combined steam and gas turbine plant,
- 4.4 Requirements of combustion chamber, types of combustion chambers.

5 Jet Propulsion **02 Hours** **04%**

- 5.1 Fundamental of propulsion technology, Turbojet Engine, thrust, thrust power, propulsive efficiency, thermal efficiency, Turboprop, Ramjet and Pulsejet engines

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behaviour will be observed strictly.
- Industrial visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of the assignment.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Evaluate performance of steam turbine

CO2: Analyze the performance of gas turbine.

CO3: Illustrate the working principle of different types of Jet propulsive engines and Rockets.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	3	-	-	2
CO2	2	-	-	2
CO3	2	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E Recommended Study Material:**Text Books/ Reference Book:**

1. P.K. Nag, “ Power Plant Engineering”, McGraw-Hill Education
2. R. K. Hegde, “ Power Plant Engineering”, Pearson India Education
3. V. Ganeshan, “Gas Turbines”, McGraw Hill Education
4. William J. Kearton, “Steam Turbine Theory and Practice”, CBS Publication
5. P. Shlyakhin, “Steam Turbines: Theory and Design”
6. Lee, “Theory and design of Steam and Gas Turbine”, McGraw Hill.
7. W.J.Kearton, “Steam Turbine Theory and Practice”

Reading Materials, web materials with full citations:

1. <http://nptel.ac.in>
2. <http://www.sciencedirect.com/science/journal>
3. <http://www.ias.ac.in/sadhana>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUE507 COMPUTATIONAL FLUID DYNAMICS & HEAT TRANSFER

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	0	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction	04
2.	Mathematical Behavior of Partial Differential Equations	04
3.	Finite Difference Technique	10
4.	Finite Element Method	10
5.	Finite Volume Method	06
6.	Numeric Grid Generation	06
7.	Navier-Stokes Equations	05

Total Hours (Theory): 45

Total Hours (Lab): 00

Total Hours: 45

B. Detailed Syllabus:

1. Introduction **04 Hours** **09%**

1.1 Introduction of CFD. Applications of CFD.

1.2 Introduction of Software's for CFD.

1.3 Conservation equation; mass; momentum and energy equations; convective forms of the equations and general description.

2 Mathematical Behavior of Partial Differential Equations 04 Hours 09%

- 2.1 Introduction, Classification of quasi-linear partial differential equations.
- 2.2 Impact of different classes of Partial differential equations (Hyperbolic, parabolic, elliptic equations) on CFD.

3 Numerical Techniques for Discretization 10 Hours 22%

- 3.1 Introduction to Finite Differences, Taylor's expansion series, approximation techniques for first order derivative, second order derivative and mixed derivative for uniform and non-uniform grid, using forward difference scheme, backward difference scheme and central difference scheme, Steady state one-dimensional conduction equation.
- 3.2 Discretization, Steady state one dimensional conduction and convection equation analytical solution and numerical solution using forward difference, backward difference and the central difference schemes
- 3.3 Discretization of time dependent parameter using explicit scheme, implicit scheme, Discretization of governing equations, approximations in non-uniform grids, boundary conditions implementation, errors.

4 Finite Element Method 10 Hours 22%

Introduction to Finite Element Method: Basics of finite element method, stiffness matrix, isoperimetric elements, formulation of finite elements for one and two dimensional elements; applications.

5 Finite Volume Method 06 Hours 13%

Introduction to Finite Volume Method: Integral approach, Discretization and higher order schemes, Application to Complex Geometry.

6 Numeric Grid Generation 06 Hours 13%

Numerical grid generation; basic ideas; transformation and mapping

7 Navier-Stokes Equations 05 Hours 10%

Explicit and implicit methods; SIMPLE type methods; fractional step methods.

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.

- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Describe various governing equations applicable for viscous fluid flows.

CO2: Describe the mathematical behavior of partial differential equations.

CO3: Apply Finite Difference Method to one dimensional and two-dimensional Engineering problems.

CO4: Formulate and solve fluid flows and Heat transfer problems by Finite Element Method and Finite Volume Method.

CO5: Simulate the Fluid flow and heat transfer problem using analysis software.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	1	-	-	-
CO2	1	-	-	1
CO3	2	1	-	3
CO4	2	1	-	3
CO5	3	2	1	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:Reference Books:

1. Patankar, “Numerical heat transfer and Fluid Flow”, Mc.GrawHill. 2002.
2. John D Anderson Jr., “Computational Fluid Dynamics: The Basics with Applications”, McGraw-Hill, 1995.
3. Chandrupatla and Belegundu, “Introduction to Finite Element Methods”, Prentice Hall of India.

4. Peric and Ferziger, “Computational Methods for Fluid Dynamics”, Springer Publication.
5. Veersteeg and Malalasekara, “CFD: The Finite Volume Method by”, Prentice Hall, 1996.

Reference Books:

1. Carnahan B, “Applied numerical method” John Wiley and Sons-2001.
2. Dante A W, “Introduction to Computational Fluid Dynamics”, Cambridge Uni. Press, 2005.
3. Reddy J N and Gartling D K, “The Finite Element Method in Heat Transfer and Fluid Dynamics”, CRC Press, 2000.
4. White F M, “Viscous Fluid Flow”, McGraw Hill, 1991.
5. Schlichting H, Gersten K, “Boundary-Layer Theory”, 8th edition, 2004.
6. Anderson, J D, “Modern Compressible Flow: With Historical Perspective”, McGrawHill, 2002.
7. Murlidhar K and Sunderrajan T, “Computational Fluid Mechanics and Heat Transfer”, Narosa Publishing House.
8. Goshdastidar, “Computer Simulation of Flow and Heat Transfer”, Tata-McGraw Hill.

Web Materials:

1. <http://www.nptel.ac.in>

Other Materials:

1. Software’s: ANSYS, COMSOL etc.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
 CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUE508 MODELLING OF IC ENGINES

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	0	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction	02
2.	Fuel air cycles, actual cycles and their analysis	05
3.	Combustion in S.I and C.I. Engines	06
4.	Introduction to modeling	06
5.	Modelling of IC Engines	16
6.	Simulation of IC Engines	10

Total Hours (Theory): 45

Total Hours (Lab): 00

Total Hours: 45

B. Detailed Syllabus:

- 1. Introduction** **02Hours 05%**
 - 1.1 Applications, actual working of IC engines, valve and port timing diagrams. Otto, diesel and Dual cycles.
- 2 Fuel air cycles, actual cycles and their analysis** **05 11%**

Hours

 - 2.1 Reasons for variation of specific heats of gases change of internal energy and enthalpy during a process with variable specific heats, isentropic expansion with variable specific heats.
 - 2.2 Effect of variable specific heats on air standard cycles of Otto and diesel cycles.

- 2.3 Factors considered and assumptions made for fuel–air cycles, dissociation, comparison of air standard and fuel air cycles,
- 2.4 Effect of operating variables on cycle analysis, difference between actual cycle and fuel air cycle for SI and CI engines.

3 Combustion in S.I and C.I. Engines **06** **13%**
Hours

- 3.1 Stages of combustion, ignition lag and the factors affecting the ignition lag, flame propagation and factors affecting flame propagation
- 3.2 Abnormal combustion and knocking in SI engines, factors affecting knocking, effects of knocking, control of knocking, combustion chambers for S.I. engines
- 3.3 Stages of combustion, delay period /ignition lag and the factors affecting it, detonation in C.I. engines,
- 3.4 Factors affecting detonation, controlling detonation, combustion chambers for C.I. engines.

4 Introduction to modelling **06** **13%**
Hours

- 4.1 Importance of modeling; Spray equation model; Thin and thick spray model; Droplet turbulence interactions; Droplet impingement on walls.

5 Modelling of IC Engines **16** **35%**
Hours

- 5.1 Classifications; zero dimensional modeling; quasi dimensional modeling, Comparison of different combustion systems; Combustion efficiency.
- 5.2 Heat of reaction - adiabatic, constant volume combustion, constant pressure combustion, temperature drop due to fuel vaporization, adiabatic flame temperature, mean effective pressure, torque and thermal efficiency at full throttle, part throttle and supercharged conditions., flow models and combustion models.

6 Simulation of IC Engines **10** **22%**
Hours

- 6.1 SI & CI engine simulation – air standard cycle, fuel-air cycle, progressive combustion cycle and actual cycle simulation – part throttle, full throttle and supercharged conditions.

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Elaborate the fundamentals and elements of IC Engines.

CO2: Conversant with Basic Concept of Modelling

CO3: Develop modeling of IC Engine.

CO4: Use of simulation techniques for IC engines.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	-	-	-
CO2	2	1	-	-
CO3	3	1	-	1
CO4	3	-	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:**Text Books:**

1. J.I Ramos, “Internal Combustion Engine Modeling” Hemisphere Publishing Corporation, 1989.
2. James N Mattavi and Charles A Amann, “Combustion Modeling in Reciprocating Engines”, Plenum Press-1980.
3. Pkandylas G C Koltsakis and A M Stamatelos, “Mathematical modeling of Precious metals catalytic converters for diesel Nox reduction”, Proc. Institution of Mechanical Engineers.
4. Ganesan V, “Computer Simulation of spark ignition engine process“, Universities Press (I) Ltd, Hyderabad, 2001.

Reference Books:

1. Heywood J B, “Internal Combustion Engine Fundamentals” McGraw Hill Book Co., USA – 2001.

2. Ramoss A L, "Modeling of Internal Combustion Engines Processes", McGraw Hill Publishing Co., 1992.
3. Ashley Campbel, "Thermodynamic analysis of combustion engines", John Wiley & Sons, New York, 1986.
4. Benson R S, whitehouse.N.D, "Internal Combustion Engines", Paragon Press, oxford, 1979.

Web Materials:

1. <http://www.nptel.ac.in>

Other Materials:

1. www.sciencedirect.com:
2. SADHNA (Engineering Science): <http://www.ias.ac.in/sadhana/>

MEME501: Fundamentals of Combustion

Description:

This course OCME1001 –Fundamentals of Combustion is offered from SWAYAM as noc23_me27-Fundamentals of Combustion.

URL: https://onlinecourses.nptel.ac.in/noc23_me27/preview

Credit and Week:

Teaching Scheme	Week	Theory	Practical	Total
Marks	12	100	-	100
Credit	-	3	-	3

About this course:

To enable students to apply the knowledge of thermodynamics to combustion. To emphasize the basics of fuels, stoichiometry, chemical kinetics and equilibrium, mass transfer, and different types of combustion process. To explain the mathematics involved in transport processes of a reactive flow, simplifications involved and the parameters affecting different types of combustion processes.

Industry Support:

TATA steels, Thermax, BHEL

Course Layout:

Week 1: Fuels and their properties

Week 2: Review of basic thermodynamics of ideal gas mixtures

Week 3: Stoichiometry

Week 4: First and Second Laws of Thermodynamics applied to combustion; Heat, temperature and composition of products in equilibrium

Week 5: Mass transfer basics

Week 6: Fundamentals of combustion kinetics

Week 7: Governing equations for a reacting flow

Week 8: General characteristics of combustion flame and detonation

Week 9: Laminar flame propagation-Flammability limits and quenching of laminar flames-Ignition-Flame stabilization

Week 10: Gas jets and combustion of gaseous fuel jets

Week 11: Turbulent premixed and non-premixed flames

Week 12: Droplet evaporation and combustion

Week 13: Combustion of a carbon particle.

Books and references:

1. S. R. Turns, An Introduction to Combustion Concepts and Applications, McGraw Hill, 2010
2. F. A. Williams, Combustion Theory, ABP
3. I. Glassman, Combustion, Academic Press
4. K. Kuo, Principles of Combustion, John Wiley

CRITERIA TO GET A CERTIFICATE

Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course .

Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.

Certificate will have your name, photograph and the score in the final exam with the breakup. It will have the logos of NPTEL and IIT Madras. It will be e-verifiable at nptel.ac.in/noc.

Only the e-certificate will be made available. Hard copies will not be dispatched.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
 CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
HSUA501: ACADEMIC WRITING

Credits and Schemes:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Academic Writing and Research Process	05
2.	Anatomy of Academic Writing	05
3.	Key Academic Skills	05
4.	Accuracy in Academic Writing	05
5.	Using and Citing Sources of Ideas	05
6.	Contemporary Practices in Academic Writing	05

Total hours (Theory): 00

Total hours (Practical): 30

Total hours: 30

B. Detailed Syllabus:

1.	Academic Writing and Research Process	5 Hours
	Introduction to Academic Writing, Academic Writing as a Part of Research, Types of Academic Writing, Features of Academic Writing, Importance of Good Academic Writing in various Academic Works	
2.	Anatomy of Academic Writing	5 Hours
	Academic Vocabulary, Simple and Complex Sentences, Organizing Paragraphs, The Writing Process, Adopting Academic Writing Style	
3.	Key Academic Skills	5 Hours
	Note – taking, Note – making, Paraphrasing, Summarizing	
4.	Accuracy in Academic Writing	5 Hours
	Lexical Range, Academic Language and Structures, Elements of Writing, Proof Reading, Editing, and Rewriting	
5.	Using and Citing Sources of Ideas	5 Hours

	Academic Texts and their Types, Intellectual Honesty in Academic Writing, Avoiding Plagiarism – Idea Theft, Degrees of Plagiarism, Types of Borrowing, Anatomy of Citations, Common Citation Styles	
6.	Contemporary Practices in Academic Writing	5 Hours
	Analytical Essays, Graph / Table / Process Interpretation and Description, Writing Reports and Abstract, Writing Research / Concept Papers	

C. Instructional Method and Pedagogy:

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like writing, group work, independent and collaborative research, etc.

D. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sr. No.	Component	Number	Marks per incidence	Total Marks
1	Paragraph Writing	1	3	03
2	Note-taking / Note-making	1	3	03
3	Paraphrasing / Summarizing	1	4	04
4	Essay Writing	1	5	05
5	Concept Paper Writing	1	10	10
5	Attendance and Class Participation			05
Total				30

External Evaluation

The University Practical Examination will be for 70 marks and will test the professional communication skills and academic writing.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical /Quiz/ Project / Academic Writing	-	70	70
Total				70

E. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Explain the concept and applications of academic writing

CO2: Acquire enough knowledge of academic writing style, strategy and approach.

CO3: Demonstrate error free and effective academic writing.

CO4: Demonstrate ability to work on project/report/paper writing.

CO5: Understand the Research and Research Methodology.

CO6: Effectively communicate in diverse academic and professional settings.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	1	3	1	1
CO2	-	3	-	-
CO3	-	3	-	-
CO4	-	3	-	-
CO5	3	3	-	-
CO6	-	3		

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

F. Recommended Study Material:

Essential Reading for Concepts

- Academic Writing for International Students, Routledge
- Academic Writing: A Guide for Management Students and Researchers. Monipally, M. M. & Pawar, B. S. Sage. 2010. New Delhi

Essential Reading for Activity and Teacher Resource

- *Effective Academic Writing Level - 1,2,3,4 (Second Edition)* By: Alice Savage, Patricia Mayer, Masoud Shafiei, Rhonda Liss, & Jason Davis; Publisher: Oxford

Additional Reading

- Writing Your Thesis (2nd Edition) by Paul Oliver, Sage
- Development Communication In Practice by Vilanilam V J, Sage
- Intercultural Communication by Mingsheng Li, Patel Fay, Sage
- www.owl.perdue.edu

M. Tech. (Thermal Engineering) Programme

SYLLABI **(Semester –3)**

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUP601 : PROJECT PRELIMINARIES

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	-	-	4
Marks	-	-	100	

A Outline of the Course:

- Project Preliminaries include course work on a specialized Subject or a Seminar.
- The course work shall be related to the area of his/her project research work.
- The coursework may be chosen from the existing PG (M. Tech.) Programmes of the registering department or from those of other departments.
- In addition to existing courses of the M. Tech. Programme, a department may offer special courses or seminar topics to the students.
- The specific subject of course work study will be decided by the department level Post Graduate Committee on recommendation of the supervisor(s).
- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Resident students will satisfy the course requirement by attending Institute and Departmental courses, and submitting technical writings on assigned course.
- Non-resident students (full time research in R&D institutions/industries) will submit equivalent amount of technical writing and reports on seminars from their own places of work.
- Course offered at 3rd Semester will be evaluated at least once during the semester and at the end of the semester as a part of continuous evaluation.
- The course work and report is expected to show clarity of thought and expression, critical appreciation of existing literature and analytical or experiment or software based applications in the field of specialization.
- Course work can be based on topic of the Project Work. It may include literature review, required theoretical input, study and comparison of various approaches for the proposed project work.
- A student has to produce some useful outcome by conducting experiments or simulations.

- Student can learn all aspects and functionality of specialized software in the institute.
- Student can generate one working test bench using research paper along with results verified from referred paper through simulation and by modify existing work student can produce one research paper (research studies)

B Instructional Methods and Pedagogy:

- Students will select related topic based on subjects they learnt and other literatures like books, periodicals, journals and various internet resources.
- Students can select the topic based on the research areas of available supervisor/guide.
- Each student has to prepare a write-up of about 30-40 pages in prescribed format only. The report typed on A4 sized sheets and bound should be submitted after approval by the guide and endorsement of the Head of Department.
- Performance of students at the seminar will be assessed by the department level Post Graduate Committee. The supervisor/guide based on the quality of work and preparation and understanding of the candidate shall do an assessment of the course.

C Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Explore current issues and research areas.

CO2: Learn the additional technical skill useful for the dissertation work.

CO3: Explain various research methodologies

CO4: Develop technical writing and communication skill.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	-	1	-
CO2	3	-	-	1
CO3	2	-	-	2
CO4	-	3	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D Recommended Study Material:

1. Programming Languages: C, C⁺⁺, MATLAB
2. Softwares: AutoCAD, Creo, ANSYS, COMSOL
3. IEEE : www.ieeexplore.ieee.org

4. www.sciencedirect.com
5. Sadhna (Engineering Science) (<http://www.ias.ac.in/sadhana/>)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUP602 : PROJECT PHASE I

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	-	-	18
Marks	-	-	500	

A Outline of the Course:

- The Project shall be related to the major field of his/her PG specialization work.
- The Project should be one of the major pieces of evidence that students are familiar with or that student wants to be familiar with. It should reflect specialist subject by means of deep and sustained study.
- The project will be finalized by the department level Post Graduate Committee on recommendation of the supervisor(s).
- The project work shall be carried out by each candidate independently during the third and fourth semester under the guidance of one of the faculty members of the Department. If the project work is of inter-disciplinary nature, a co-guide shall be taken from the same or any other relevant Department.
- If a project work has to be carried out in any industry / factory / organization, outside the campus, the permission to that effect and the name of co-guide at any of these organizations shall be intimated to the Post Graduate Committee at the beginning of third semester.
- Project I includes literature review, required theoretical input, study and comparison of various approaches for the proposed dissertation work.

B Instructional Methods and Pedagogy:

- Student has to submit a project/dissertation proposal indicating the tentative title and broad outline of the proposed work and the name(s) of the supervisor(s) along-with their concurrence in writing within 30 days from the starting of the third semester.
- Utmost care should be taken in selection of research topic so that repetition of research work is avoided.

- Project - I will be evaluated at least once during the semester and at the end of the semester as a part of continuous evaluation.
- After successful completion of Project I only students are allowed to go register for Project – II.

C Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Apply concept to formulate the research problem.

CO2: Identify the research area by critical literature review and establish scope of work.

CO3: Develop study / experimental methodology.

CO4: Develop technical writing and communication skill.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	3	-	-	3
CO2	2	-	2	2
CO3	3	-	-	3
CO4	-	3	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D Recommended Study Material:

- 1 Programming Languages: C, C⁺⁺, MATLAB
- 2 Softwares: AutoCAD, Creo, ANSYS, COMSOL
- 3 IEEE : www.ieeexplore.ieee.org
- 4 www.sciencedirect.com

M. Tech. (Thermal Engineering) Programme

SYLLABI **(Semester –4)**

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUP603 : PROJECT PHASE II

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	-	-	18
Marks	-	-	500	

A Outline of the Course:

- Student should carry out the investigation by identifying sources of evidence, accessing those using accepted and rigorous academic methods, and analysing and interpreting the material gathered by simulation/experimentation.
- A project - II is student's own work and will need to keep up the effort, and the interest, over several months and through several stages.
- Student need to think carefully about the time necessary to carry-out and complete your project work and the relative writing up.
- The project should present an orderly and critical exposition of the existing knowledge of the subject and will embody results of original investigations demonstrating the capacity of the candidate to do independent research work.
- While writing the thesis/dissertation, the candidate will layout clearly the work done by him independently and the sources from which he has obtained other information contained in his/her Dissertation

B Instructional Methods and Pedagogy:

- Project - II will be evaluated at least once during the semester and at the end of the semester as a part of continuous evaluation.
- Before submission of Phase II project/dissertation report, it is expected from a student to publish at least one research paper in National/International conference. Further, for such

publications, Department Post Graduate Committee will identify and approve the national/international conferences.

- The dissertation shall be submitted for ‘dissertation – evaluation’ ordinarily at the end of IV Semester and ‘dissertation – open defense’ shall be held soon after the submission of the dissertation.

C Student Learning Outcomes:

At the end of the course, the students will be able to

CO1: Design and create the experiment setup to collect data.

CO2: Analyse and conclude experimental results.

CO3: Write and publish the technical research paper.

CO4: Develop technical writing and communication skill.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	3	-	2	3
CO2	3	-	-	3
CO3	-	3	1	1
CO4	-	3	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D Recommended Study Material:

- 1 Programming Languages: C, C⁺⁺, MATLAB
- 2 Softwares: AutoCAD, Creo, ANSYS, COMSOL
- 3 IEEE : www.ieeexplore.ieee.org